AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 1: Scoping, architecture and real data acquisition

Mon 2 Nov - Fri 6 Nov 2026 · Sprint 1 · Overall progress 10%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 6 Nov 2026

On track **This week in one line:** Fixed the project scope, chose two real battery-ageing datasets (NASA PCoE and CALCE CS2), designed the layered system architecture and set up the repository.

1. Goal for the week

Fix the problem statement and scope, find real battery data that covers the whole battery life, and design the system before writing models.

2. Day-by-day work

Day	Work done
Mon 2 Nov	Wrote the problem statement and five objectives: SOH estimation, RUL prediction, second-life grading, Digital Battery Passport, analytics interface.
Tue 3 Nov	Surveyed public Li-ion ageing datasets (NASA PCoE, CALCE, Oxford, MIT/Toyota) for size, signals and how far each cell is aged.
Wed 4 Nov	Chose NASA PCoE (impedance and temperature) plus CALCE CS2 (aged down to 15-27% SOH, so it covers second life). Downloaded both.
Thu 5 Nov	Studied the EU Battery Regulation 2023/1542: passport mandatory from 18 Feb 2027 (Art. 77, Annex XIII), access tiers, battery status values.
Fri 6 Nov	Designed the six-layer architecture, wrote config.py with all thresholds, created the repository and this report.

3. Completed deliverables

- **batterylife/config.py**: datasets, lifecycle thresholds (80% / 60% SOH), second-life application catalogue, sustainability assumptions.
- **scripts/download_data.py**: fetches the raw data.
- **docs/architecture.md** and the architecture diagram (below).

4. Results

Decision	Choice	Reason
Datasets	NASA PCoE (4 cells) + CALCE CS2 (4 cells)	Real data; together they cover first life, second life and recycling
First-life end	80% SOH	Common EV retirement threshold
Second-life floor	60% SOH	Below this re-use is not worthwhile
SOH inputs	Charging signals only	A battery management system sees them every charge
Passport	Signed, hash-chained events	Aligned with EU Regulation 2023/1542, Annex XIII

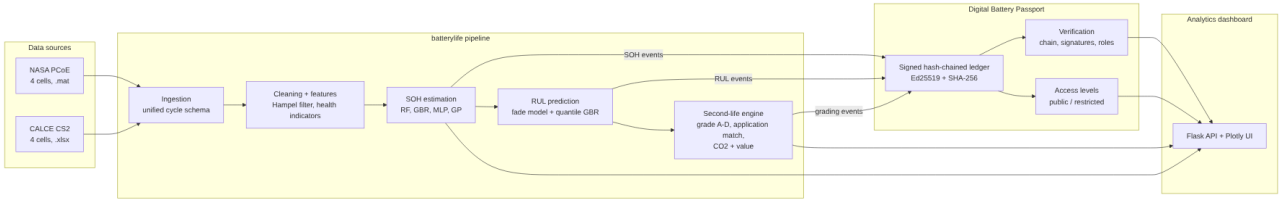


Figure 1. System architecture: data sources, pipeline, Digital Battery Passport and dashboard.

5. Problems faced and how they were solved

Problem	Resolution
Only 8 cells exist in total across both datasets.	Validate with leave-one-cell-out and report errors per cell.
CALCE raw data is about 170 MB.	Kept out of git; a download script fetches it.

6. Plan for next week

- Parse both datasets into one validated cycle-level table.
- Check data quality and document the findings.

Faculty remarks

Faculty signature: _____ Student signature: _____